Individual Task 3 - Module 3

Machine learning: Concepts, Algorithms and Application

Feature Extraction Thought Experiment: Selecta dataset(e.g.,photos,shopping lists) and describe which features would be important to a machine learning model.

Dataset Selected: Online Shopping Transaction Data

Introduction to Feature Extraction

In the world of Machine Learning, raw data alone is not enough to build accurate predictive models. Real-world datasets, especially online shopping transaction data, are usually large, messy, and complex. They contain multiple columns, different formats, missing values, and irrelevant information.

Feature extraction is the process of selecting, transforming, and organizing important variables (features) from raw data so that a Machine Learning model can learn effectively.

A feature is any measurable property or characteristic of the data that helps in making predictions.

For example, in an online shopping dataset:

- Customer age
- Product price
- Purchase time
- Payment method
- Review sentiment

All these can act as features.

Without proper feature extraction:

- The model may learn incorrect patterns.
- Training time increases.
- Accuracy decreases.
- Overfitting may occur.

Therefore, feature extraction is one of the most critical steps in building a reliable ML system.

Description of the Online Shopping Dataset

An online shopping transaction dataset may contain:

Customer Information

- Customer ID

- Age
- Gender
- Location
- Account registration date

Product Information

- Product ID
- Category
- Price
- Brand
- Rating

Transaction Information

- Order ID
- Date and time of purchase
- Payment method
- Discount applied
- Quantity purchased

Behavioral Data

- Click history
- Search history
- Time spent on pages
- Wishlist items

Review Data

- Customer reviews (text)
- Star ratings
- Feedback comments

Such a dataset is multi-dimensional and contains:

- Numerical data
- Categorical data
- Text data
- Time-series data

Proper feature extraction helps convert this complex dataset into structured input for ML models.

Important Features and Their Detailed Explanation

A. Customer-Related Features

1. Purchase Frequency

This represents how often a customer makes purchases in a given period (weekly, monthly, yearly).

Why it matters:

- Frequent buyers are more engaged.
- They are more likely to respond to recommendations.
- Low purchase frequency may indicate churn risk.

Example:
If a customer buys products every week, the system can classify them as an active user.

2. Average Spending per Order

This calculates the average amount spent per transaction.

Why it matters:

- Helps identify premium customers.
- Useful for customer segmentation.
- Important for targeted marketing strategies.

High-spending customers may receive exclusive offers.

3. Customer Location

Location affects:

- Product demand
- Shipping cost
- Delivery time
- Regional preferences

Example:
Customers in metropolitan areas may prefer fast delivery and digital payment methods.

4. Account Age

Number of days/months since account creation.

Why it matters:

- Older accounts may indicate loyal customers.
- New accounts may require engagement strategies.

B. Product-Related Features

1. Product Price

Price strongly influences buying decisions.

Higher prices may:

- Require more reviews.
- Be more sensitive to discounts.

Lower prices may encourage impulse purchases.

2. Product Category

Categories like:

- Electronics
- Clothing
- Groceries
- Beauty products

Why it matters:

- Helps build recommendation systems.
- Identifies customer interests.

3. Discount Percentage

Represents promotional offers applied to products.

Why it matters:

- Higher discounts increase purchase probability.
- Useful for seasonal sales analysis.

4. Product Rating

Average rating given by customers.

Why it matters:

- High ratings build trust.
- Low ratings may reduce conversions.

C. Behavioral Features

1. Click History

- Number of times a product is viewed.

Why important?

- Repeated views indicate interest and help in retargeting.

2. Time Spent on Product Page

- Longer time suggests serious buying intention.
- Short time may indicate casual browsing.

3. Search Keywords

- Search terms reveal customer intent.

Example:
Searching “budget smartphone” indicates price sensitivity.

D. Text-Based Features (Reviews)

Online reviews are unstructured data and must be converted into numerical features.

1. Sentiment Score

- Classifies reviews as positive, negative, or neutral.

2. Keyword Frequency

- Identifies frequently used words in reviews.
- Highlights product strengths and weaknesses.

Feature Engineering

Feature engineering improves raw features by transforming them into more meaningful forms.

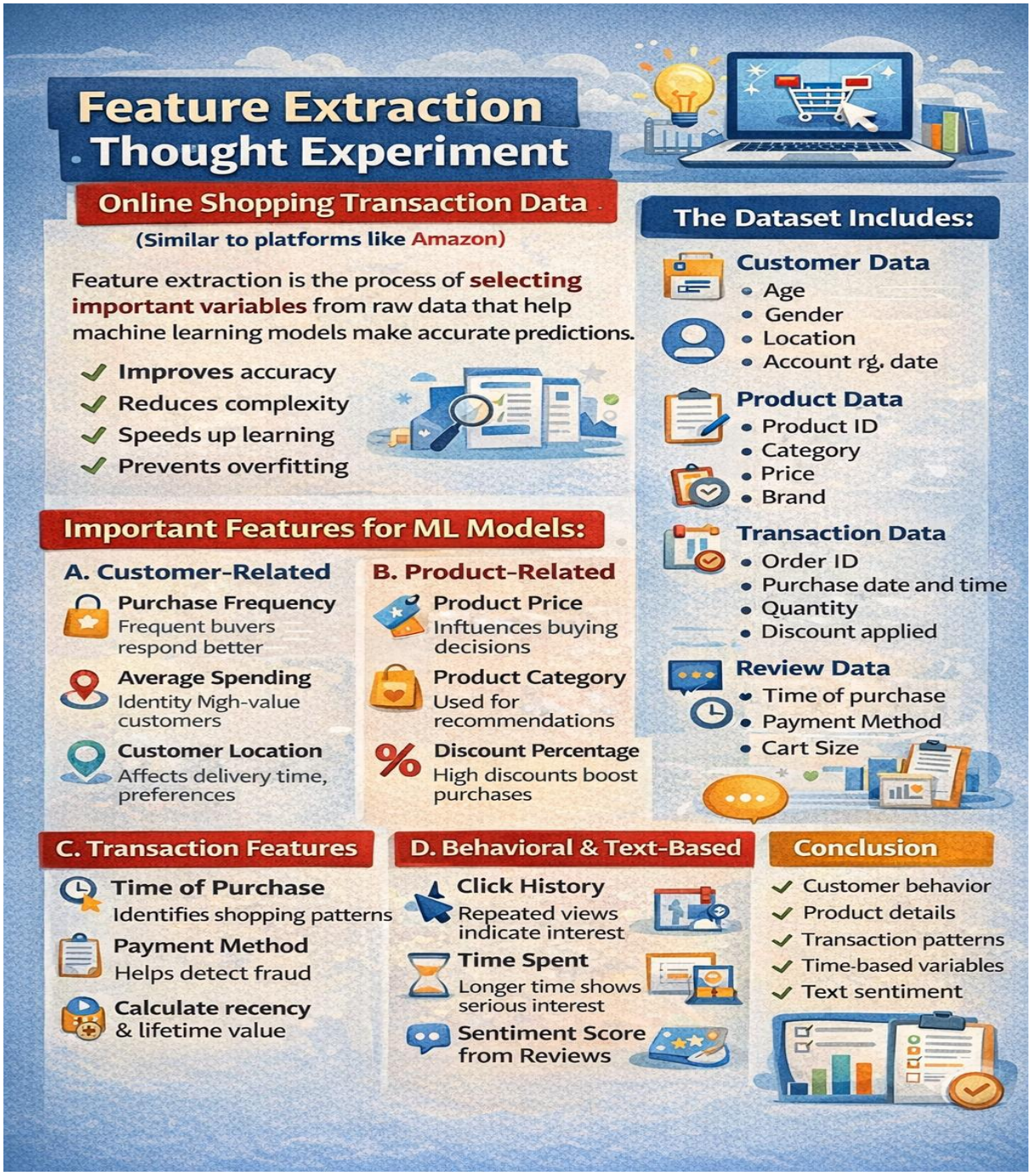
Examples:

- Convert exact age into age groups (18–25, 26–35, etc.)
- Extract weekend or weekday from timestamp
- Calculate Recency (days since last purchase)
- Create Customer Lifetime Value (CLV)
- Generate engagement score based on clicks and time spent

These engineered features often improve model performance.

Importance of Feature Extraction

Proper feature extraction:

- Improves prediction accuracy

- Reduces training time
- Prevents overfitting
- Makes models easier to interpret
- Removes irrelevant information